

Retrieval-Augmented Generation for an Automated Response System

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- **Problem**

- Recurring MUIA inquiries require significant time to answer

- **Hypotheses**

- Information in public UPM/MUIA websites
- Information in past institutional replies

- **Main challenges**

- Emails may contain sensitive information
- Website content changes at least yearly
- Difficult to separate performance from data coverage

Main objectives

- Create data pipelines for emails/websites
- Study system feasibility and cost
- Optimize performance

- **Language models**

- What advances make it possible to automate such a system?

- **Retrieval pipeline**

- What are the common stages in a retrieval pipeline?

- **Evaluation strategies**

- How can such a system be evaluated?

State of the art: Transformer-based models

- **Why the Transformer** (Vaswani et al., 2017)
 - Helps maintain long-range dependencies
 - Parallel training
- **Encoder-only models**
 - Bidirectional attention
 - Common for sentence embedding and retrieval
- **Decoder-only models**
 - Causal attention ($M_{ij} = -\infty$ for $j > i$)
 - Common for autoregressive language generation

State of the art: Embedding models for retrieval

- **Masked lang. modeling**
 - BERT (Devlin et al., 2019)
- **Bi-encoder repr.**
 - SBERT (Reimers and Gurevych, 2019)
- **Contrastive retrieval**
 - DPR (Karpukhin et al., 2020)
- **Multi-vector late-interact.**
 - ColBERT (Khattab and Zaharia, 2020)
- **Unified sparse/dense/late**
 - M3 (Chen et al., 2024)
- **Decoder-based**
 - Qwen3 (Zhang et al., 2025), Jina (Akram et al., 2026)

State of the art: Decoder models for generation

- **Autoregressive language modeling**

- GPT-2 (Radford et al., 2019), GPT-3 (Brown et al., 2020)

- **Instruction following**

- Pretraining → SFT → RL

- **Prompting and reasoning**

- In-context learning (ICL) (Brown et al., 2020)
- Chain-of-Thought (Wei et al., 2022)
- Programming/tool use: PAL (Gao et al., 2023), agents

- **(Context/capacity) scaling**

- GQA (Ainslie et al., 2023), RoPE (Su et al., 2024)
- Sparsely-Gated MoE (Shazeer et al., 2017)

State of the art: Retrieval pipeline main stages

- **Document cleanup, transformation, and chunking**

- Noise removal, rewriting, chunking

- **Document encoding**

- Sparse: TF-IDF (Sparck Jones, 1972), BM25 (Robertson et al., 1994), M3 sparse, SPLADE, etc.
- Dense: Qwen3 Embedding, Jina-v5-text-small, M3 dense, etc.

- **Indexing and storage**

- HNSW (Malkov and Yashunin, 2018), PQ (Jegou et al., 2010), etc.
- Embedding to chunk mapping

- **Query encoding and retrieval**

- Query expansion, transformation: HyDE (Gao et al., 2023), etc.
- Query encoding and index search (k-NN, k-ANN)

- **Fusion and reranking**

- RRF (Cormack et al., 2009): $\text{RRF}(d) = \sum_{r \in R_d} \frac{1}{k+r(d)}$
- Cross-encoder (BERT-style) reranking

- **Retrieval evaluation metrics**

- Hit@k
- Precision@k
- Recall@k
- MRR@k

- **RAG response quality evaluation**

- RAGAS (Es et al., 2024): Faithfulness, Answer Relevance, Context Relevance

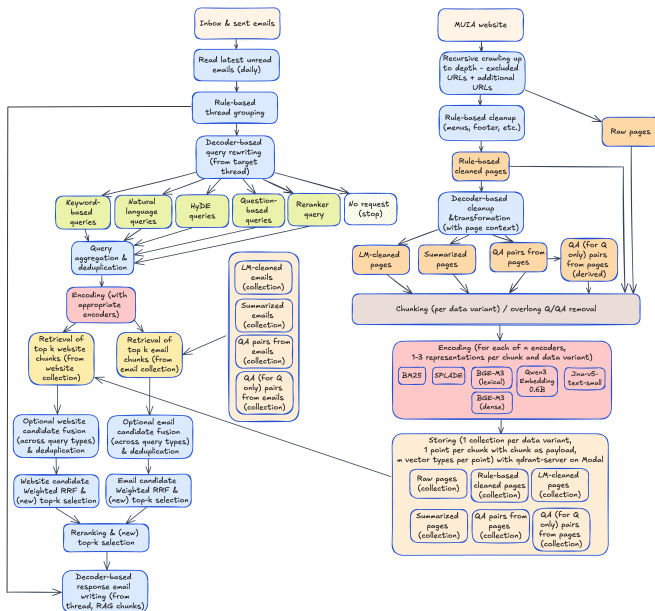
- **LLM-as-a-judge evaluation**

- LM as a human proxy (Chiang et al., 2023; Zheng et al., 2023)

• How do constraints shape design?

Constraint	Design decision
Email-data sensitivity	Open-weight + anonymization for LM-judge
Yearly updating of websites	RAG with yearly re-crawling/re-encoding
Predictable budget	Retrieval-based, non-agentic system
Recurring (e.g., daily) email inflow	Autonomous system
Uncertain best configuration	Exploration-exploitation balance

Methodology: System overview



Methodology: Website pages data pipeline

- **Start URL**
 - <https://muia.dia.fi.upm.es/es/>
- **Recursive crawling + additional/excluded URLs**
 - Max depth=3
- **Rule-based cleanup + decoder-based transformation**
- **Data variants**
 - Raw chunks
 - Rule-based cleaned chunks
 - LM cleaned text chunks
 - LM summary chunks
 - LM Q&A chunks
 - LM Q&A only Q encoded chunks
- **Chunk and encode**

Webpage category	Domain	Unique URL count
University	www.upm.es / other	109
Faculty	www.fi.upm.es	16
Department	dia.fi.upm.es	9
Master	muia.dia.fi.upm.es	48
Total		182

Methodology: Email dataset pipeline (1/2)

- **Original dataset**

- 8,183 messages across 6 folderURI

- **Dataset cleaning**

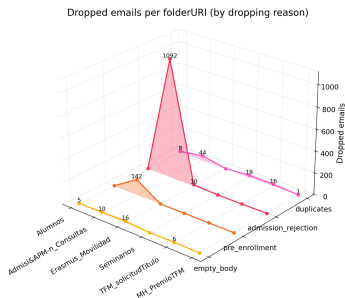
- 6,814 / 1,369 kept/discarded messages

- **Thread grouping**

- Rule-based+decoder-based: 6,510 messages from 2,713 threads

- **Filtering**

- 4,551 / 1,779 + 180 messages from 1,889 / 678 + 146 kept/discarded threads (internal+director-only)



Methodology: Email dataset pipeline (2/2)

- **Thread-based sample creation**

- 2,959 samples (inbound-based) from 1,889 threads

- **Thread-based splitting**

- train: 2,071 samples from 1,324 threads
- dev: 437 samples from 270 threads
- test: 451 samples from 295 threads

- **Thread-based KB curation**

- 1,324 threads + 6 templates → 846/484 kept/discarded threads (student-/prospect-only)
- 846 threads → 713 (useful) chunks
- 4 (anonymized) decoder-based data variants to encode:
 - LM cleaned text chunks
 - LM summary chunks
 - LM Q&A chunks
 - LM Q&A only Q encoded chunks

Methodology: M3 fine-tuning dataset preparation (1/3)

- **Query rewriting**

- 2,071 train samples $\rightarrow$ 1,542/513/16 request/no-request/exception samples
- 32 + 1 (reranker) + 1 (anonymized) queries per request sample

- **Retrieval pipeline**

- Both data sources & LM cleaned text chunks variant
- BM25 + SPLADE + M3 sparse + M3 dense + Qwen3 + Jina $\rightarrow$ Max-sim. + WRRF $\rightarrow$ reranking
- 2 sources $\times$ 2 encoder repr. (M3 dense & sparse) $\times$ top-16 + (reranker) top-5 chunks = 69 chunks

- **LLM-as-a-judge**

- Over anonymized query + 69 chunks
- Full/partial/non-answerability (1,520 full/partial)
- Supporting/insufficient chunks

Methodology: M3 fine-tuning dataset preparation (2/3)

• Query sampling

- Sample 4x, 8x or 16x queries (consistently)

Query type	Count	Weight	Expected count (exp=8x)
Keyword	8	0.068	0.54
Natural	8	0.129	1.03
HyDE	8	0.355	2.84
Question	8	0.157	1.26
Reranker	1	0.291	2.33
Total	33	1	8

• Passage sampling

- For each expanded query, sample 1 p^+ and 7 p^- :
 - 1 p^+ sampled from supporting chunks
 - 2 p^- sampled from oracle insufficient chunks (overall hard negatives)
 - 5 p^- sampled from the non-supporting M3 top-16 chunks (M3 hard negatives)

• InfoNCE-IBN collision estimation

- Topic recurrence + expansion from same request
- Monte Carlo simulation (100 runs) under expansion = 8:

Batch size	Collisions per sample		
	Positive	Pos. + Group	Pos. + Group + Similar
1	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00
2	0.02 $\pm$ 0.00	0.14 $\pm$ 0.00	0.18 $\pm$ 0.00
4	0.06 $\pm$ 0.00	0.41 $\pm$ 0.01	0.56 $\pm$ 0.01
8	0.14 $\pm$ 0.00	0.95 $\pm$ 0.01	1.45 $\pm$ 0.02
16	0.30 $\pm$ 0.01	2.03 $\pm$ 0.02	3.67 $\pm$ 0.04
32	0.62 $\pm$ 0.01	4.20 $\pm$ 0.02	9.63 $\pm$ 0.08

Methodology: M3 fine-tuning positive-passage mask

- Mask p^+ collisions in the IBN denominator

$$\begin{aligned} Z_i = & \exp(s_*(q_i, p_i^+) / \tau) \\ & + \sum_{p \in \mathcal{P}_i^-} \exp(s_*(q_i, p) / \tau) \\ & + \sum_{\substack{j=1 \\ j \neq i}}^{|\mathcal{B}_Q|} m_{i, p_j^+}^{\text{pos}} \exp(s_*(q_i, p_j^+) / \tau) \\ & + \sum_{\substack{j=1 \\ j \neq i}}^{|\mathcal{B}_Q|} \sum_{p \in \mathcal{P}_j^-} m_{i, p}^{\text{pos}} \exp(s_*(q_i, p) / \tau) \end{aligned}$$

$$m_{i, p}^{\text{pos}} = \begin{cases} 0 & \text{if } \text{tok}(p) = \text{tok}(p_i^+), \\ 1 & \text{otherwise.} \end{cases}$$

Methodology: M3 fine-tuning same expansion group mask

- Mask same-group query p^+ collisions in the IBN denominator
 - Includes positive-passage mask

$$\begin{aligned} Z_i = & \exp(s_*(q_i, p_i^+) / \tau) \\ & + \sum_{p \in \mathcal{P}_i^-} \exp(s_*(q_i, p) / \tau) \\ & + \sum_{\substack{j=1 \\ j \neq i}}^{|\mathcal{B}_Q|} m_{i, p_j^+}^{\text{group}} \exp(s_*(q_i, p_j^+) / \tau) \\ & + \sum_{\substack{j=1 \\ j \neq i}}^{|\mathcal{B}_Q|} \sum_{p \in \mathcal{P}_j^-} m_{i, p}^{\text{group}} \exp(s_*(q_i, p) / \tau) \end{aligned}$$

$$m_{i, p}^{\text{group}} = \begin{cases} 0 & \text{if } \exists k \in \{1, \dots, |\mathcal{B}_Q|\} : \\ & g(q_k) = g(q_i) \text{ and} \\ & \text{tok}(p) = \text{tok}(p_k^+), \\ 1 & \text{otherwise.} \end{cases}$$

Methodology: M3 fine-tuning similar group mask

- Mask same-group or similar-group p^+ collisions in the IBN denominator
 - Similar-group by 1, 520 sample-sample (reranker query) sim. ≥ 0

$$\begin{aligned} Z_i = & \exp(s_*(q_i, p_i^+) / \tau) \\ & + \sum_{p \in \mathcal{P}_i^-} \exp(s_*(q_i, p) / \tau) \\ & + \sum_{\substack{j=1 \\ j \neq i}}^{|\mathcal{B}_Q|} m_{i, p_j^+}^{\text{sim-group}} \exp(s_*(q_i, p_j^+) / \tau) \\ & + \sum_{\substack{j=1 \\ j \neq i}}^{|\mathcal{B}_Q|} \sum_{p \in \mathcal{P}_j^-} m_{i, p}^{\text{sim-group}} \exp(s_*(q_i, p) / \tau) \end{aligned}$$

$$m_{i, p}^{\text{sim-group}} = \begin{cases} 0 & \text{if } \exists k \in \{1, \dots, |\mathcal{B}_Q|\} : \\ & (g(q_k) = g(q_i) \text{ or } \text{score}_{\text{cross-encoder}}(q_i^r, q_k^r) \geq 0) \\ & \text{and tok}(p) = \text{tok}(p_k^+), \\ 1 & \text{otherwise.} \end{cases}$$

Evaluation: Human-LM alignment

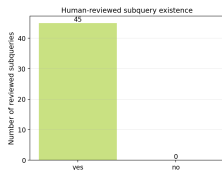
- **Query rewriter**

- 51 dev samples → 34 request samples
- ~ 88% no-request (human-LM) alignment

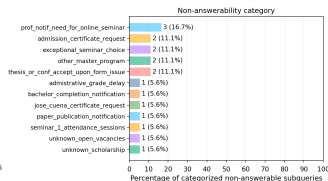
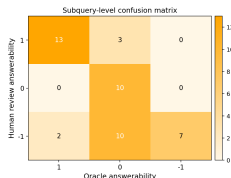
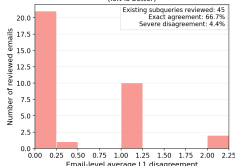
- **LLM-judge**

- Anonymized query + full LM summaries **web** collection

Human-LM judge alignment results
(human review: 2026-05-13_19-40-21)



Distribution of email-level average L1 disagreement (left is better)



General answerability and data source selection

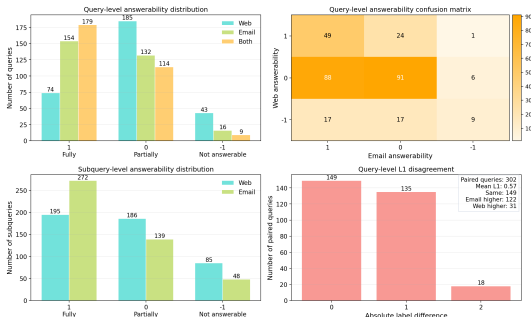
- **Query rewriter**

- 437 dev samples → 302 request samples
- 32 + 1 + 1 queries

- **LLM-judge**

- Anonymized query + full LM summaries **web** collection
- Anonymized query + full LM summaries **email** collection

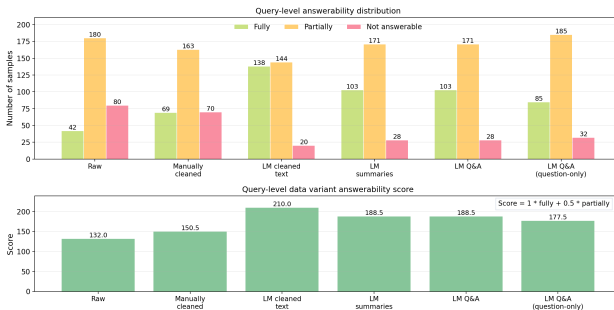
Oracle data source answerability results
(variant: lm_summary_chunks
web dump: 2026-05-12_18-05-25, web oracle: 2026-05-14_19-31-38
email dump: 2026-05-12_18-05-25, email oracle: 2026-05-15_01-52-23)



Data variant selection

- For each data variant:
 - Retrieval pipeline
 - Using both data sources, except for raw/rule-based
 - Same 32 + 1 (cached) queries → top-5 reranker chunks
 - LLM-judge
 - Anonymized query + top-5 reranker chunks

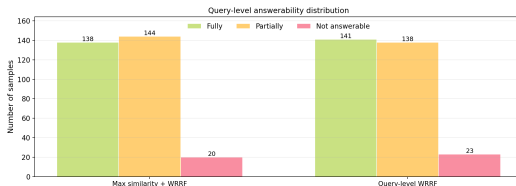
Oracle data variant answerability results
(retrieval: 2026-05-16_19-51-52
oracle: raw: 2026-05-17_01-46-08, manually cleaned: 2026-05-17_03-39-42
oracle: LM cleaned: 2026-05-17_13-48-14, LM summary: 2026-05-17_15-01-22
oracle: LM Q&A: 2026-05-17_16-11-49, LM Q&A Q-only: 2026-05-17_17-12-09)



Evaluation: Fusion strategy selection

- For LM cleaned text chunks, both sources:
 - Retrieval pipeline
 - Under max. sim. + WRRF (**existent**)
 - Under query-level WRRF (33 lists/encoder)
 - LLM-judge
 - Anonymized query + top-5 reranker chunks

Oracle fusion answerability results
(variant: lm cleaned text chunks)
max similarity + WRRF retrieval: 2026-05-16_19-51-52, oracle: 2026-05-17_13-48-14
query-level WRRF retrieval: 2026-05-18_15-33-45, oracle: 2026-05-18_17-54-42)

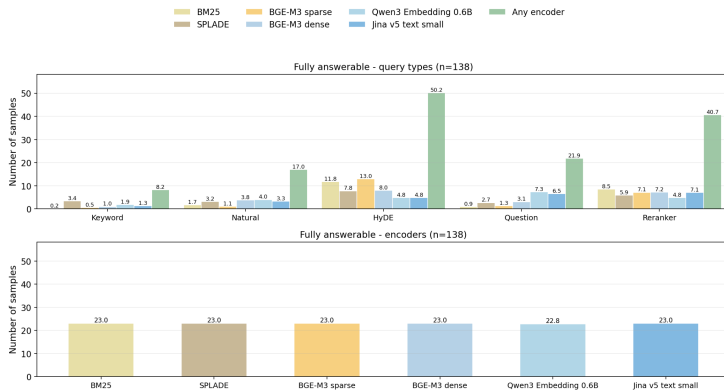


Evaluation: Encoder and query type analysis (1/2)

● Provenance under max. sim. + WRRF

- For the 138 fully answerable samples:
 - 1 point/sample $\rightarrow n$ supporting chunks $\rightarrow m$ query types or encoders/chunk

Query-type and encoder provenance of oracle-supported chunks
(fusion: Maximum similarity + WRRF, variant: Im_cleaned_text_chunks
retrieval: 2026-05-16_19-51-52, oracle: 2026-05-17_13-48-14)



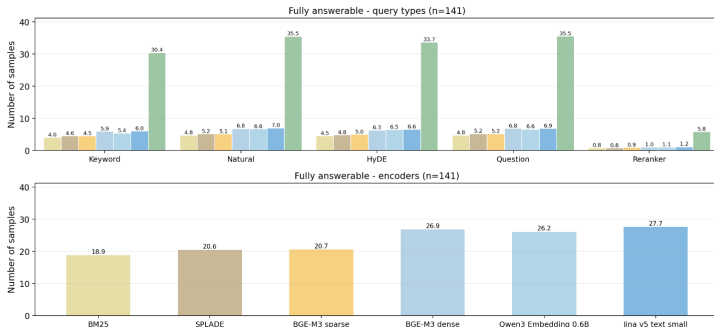
Evaluation: Encoder and query type analysis (2/2)

● Provenance under query-level WRRF

- For the 141 fully answerable samples:
 - 1 point/sample $\rightarrow n$ supporting chunks $\rightarrow m$ query types or encoders/chunk

Query-type and encoder provenance of oracle-supported chunks
(fusion: Query-level RRF, variant: lm_cleaned_text_chunks
retrieval: 2026-05-18_15-33-45, oracle: 2026-05-18_17-54-42)

BM25 BGE-M3 sparse Qwen3 Embedding 0.6B Any encoder
SPLADE BGE-M3 dense Jina v5 text small

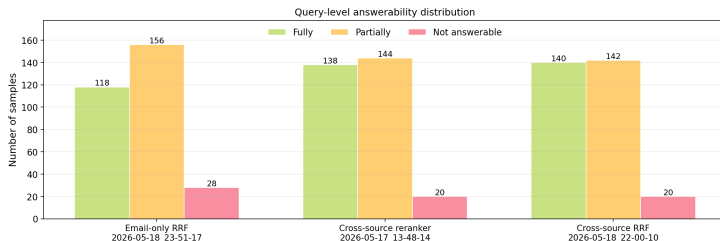


Evaluation: Reranking evaluation

• LLM-judge

- Anonymized query + top-5 reranker chunks (**existent**)
- Anonymized query + top-5 email chunks
- Anonymized query + top-5 web + top-5 email chunks

Oracle final answerability results
(variant: lm_cleaned_text_chunks)
email-only RRF oracle: 2026-05-18_23-51-17, cross-source reranker oracle: 2026-05-17_13-48-14,
cross-source RRF oracle: 2026-05-18_22-00-10)



Evaluation: Fine-tuning evaluation (1/2)

- Supporting/insufficient **dev** chunks → pairwise accuracy
 - Avoids re-encoding full collections after each SFT run
- Batch size = 8 and collision masking (enabled) works best

Batch size	Filter disabled			Filter enabled		
	Dense pairwise acc.	Sparse pairwise acc.	Dense + sparse pairwise acc.	Dense pairwise acc.	Sparse pairwise acc.	Dense + sparse pairwise acc.
4	0.6847	0.5816	0.6665	0.6828	0.5863	0.6629
	(+16.3%)	(-3.3%)	(+9.3%)	(+16.0%)	(-2.5%)	(+8.7%)
8	0.6712	0.5795	0.6555	0.6949	0.6045	0.6806
	(+14.0%)	(-3.6%)	(+7.5%)	(+18.1%)	(+0.5%)	(+11.6%)
16	0.6590	0.5878	0.6437	0.6855	0.6016	0.6737
	(+12.0%)	(-2.2%)	(+5.5%)	(+16.5%)	(+0.1%)	(+10.4%)

Evaluation: Fine-tuning evaluation (2/2)

- 15 p^- (7 hard, 8 IBN negatives) works best

Batch size	Max. effective IBN size	Dense pairwise acc.	Sparse pairwise acc.	Dense + sparse pairwise acc.
4	16	0.6768 (+15.0%)	0.6102 (+1.5%)	0.6879 (+12.8%)
8	32	0.6949 (+18.1%)	0.6045 (+0.5%)	0.6806 (+11.6%)
8	16	0.7100 (+20.6%)	0.6181 (+2.8%)	0.6996 (+14.7%)
8	12	0.6844 (+16.3%)	0.6047 (+0.6%)	0.6679 (+9.5%)

- 8x dataset query expansion works best

Expansion factor	Dense pairwise acc.	Sparse pairwise acc.	Dense + sparse pairwise acc.
4x	0.6671 (+13.3%)	0.6061 (+0.8%)	0.6635 (+8.8%)
8x	0.7100 (+20.6%)	0.6181 (+2.8%)	0.6996 (+14.7%)
16x	0.6637 (+12.8%)	0.6391 (+6.3%)	0.6803 (+11.5%)

Evaluation: Fine-tuning ablation study

- Using hard negatives **improves** over sampling all IBN p^-

Denominator sampling	Dense pairwise acc.	Sparse pairwise acc.	Dense + sparse pairwise acc.
Keep query negatives, sample remaining	0.7100 (+20.6%)	0.6181 (+2.8%)	0.6996 (+14.7%)
Sample all negatives	0.6700 (+13.8%)	0.5997 (-0.3%)	0.6584 (+7.9%)

- Using all three masks performs **best**

Filtering configuration	Dense pairwise acc.	Sparse pairwise acc.	Dense + sparse pairwise acc.
Group + similar-group filtering	0.7100 (+20.6%)	0.6181 (+2.8%)	0.6996 (+14.7%)
Group filtering only	0.6904 (+17.3%)	0.6024 (+0.2%)	0.6799 (+11.5%)
Duplicate-positive filtering only	0.6988 (+18.7%)	0.6001 (-0.2%)	0.6742 (+10.5%)

Evaluation: Post-SFT M3 performance

- **Gather top-5 per encoder repr.**
 - 2 data sources $\times$ 6 encoder repr. $\times$ top-5 chunks = 60 chunks
- **LLM-judge**
 - Anonymized query + 60 chunks **pre-SFT**
 - Anonymized query + 60 chunks **post-SFT**
- M3 (dense: +54%) and single encoder performance improves

Bundle size	Pre-SFT			Post-SFT		
	Full score	Partial score	Aggr. score	Full score	Partial score	Aggr. score
1	90.7	49.2	115.3	123.8 (+36.5%)	62.5 (+27.0%)	155.1 (+34.5%)
2	134.5	76.9	172.9	156.6 (+16.4%)	74.8 (-2.7%)	194.0 (+12.2%)
3	152.3	91.0	197.8	172.1 (+13.0%)	84.6 (-7.0%)	214.4 (+8.4%)
4	162.7	101.1	213.2	182.6 (+12.2%)	94.5 (-6.5%)	229.9 (+7.8%)
5	170.6	107.2	224.2	187.8 (+10.1%)	99.7 (-7.0%)	237.7 (+6.0%)
6	176.0	114.0	233.0	191.0 (+8.5%)	102.0 (-10.5%)	242.0 (+3.9%)

Evaluation: Post-SFT WRRF weight rebalancing

- M3 needs stronger voting to improve full-ensemble performance
 - M3 dense: 0.577
 - M3 sparse: 0.236

Oracle post-SFT final answerability results

(variant: lm_cleaned_text_chunks)

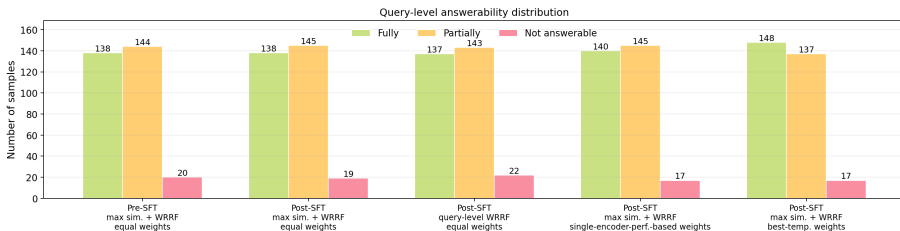
Pre-SFT max sim. + WRRF equal weights oracle: 2026-05-17_13-48-14

Post-SFT max sim. + WRRF equal weights oracle: 2026-06-16_20-38-47

Post-SFT query-level WRRF equal weights oracle: 2026-06-28_19-46-17

Post-SFT max sim. + WRRF single-encoder-perf.-based weights oracle: 2026-06-17_10-26-13

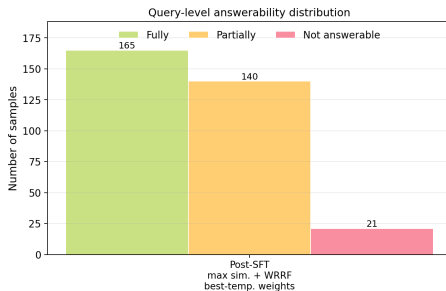
Post-SFT max sim. + WRRF best-temp. weights oracle: 2026-06-29_21-40-27)



Evaluation: Test set performance

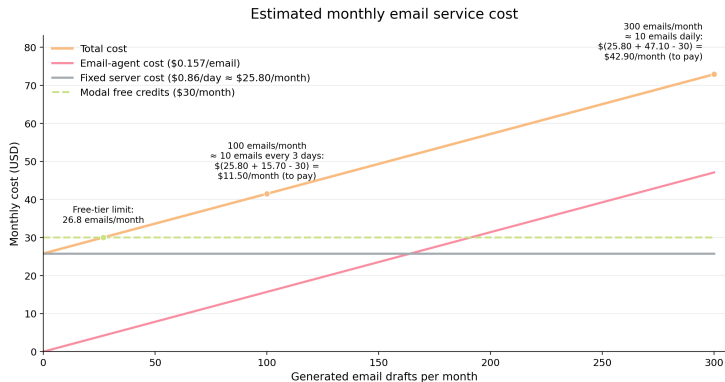
- Test set improves dev set performance

Oracle test answerability results
(variant: lm_cleaned_text_chunks
split: test, oracle: 2026-06-30_15-01-07)



Evaluation: Production cost estimate

- Re-crawling/re-encoding: \$120/year
- Email writing: variable
 - 26.8 emails/month → Free
 - 100 emails/month → \$11.50
 - 300 emails/month → \$42.90



Conclusions

- MUIA email autoresponding system is feasible
- Importance of exploration and data source, data variant, encoder complementarity, encoder fine-tuning, etc. on performance
- SFT successfully adapts encoder to MUIA's data
- Using all three masks performs best

Main directions for future work

- Periodic email pipeline (e.g., yearly)
- More data for encoder fine-tuning to simplify full retrieval pipeline
- Increase context (60 chunks $\rightarrow$ 191 fully ans.) / model strength
- Serve from UPM's servers

Example (1/2)

Incoming email — Schedule query

Hola,

¿Cuándo se imparte Redes Bayesianas? No encuentro el horario en Moodle.

Gracias.

Example (1/2)

Generated draft — Schedule query

Hola,

El horario para la asignatura de Redes Bayesianas (A4) en el curso 2025–26 es el siguiente:

- Martes: 09:00–10:00
- Miércoles: 09:00–10:00
- Jueves: 17:00–19:00

Ten en cuenta que esta asignatura se imparte de forma concentrada durante las primeras ocho semanas del semestre.

Saludos,
Damiano Zanardini

Example (2/2)

Incoming email — Fees and scholarships

Dear Damiano,

Could you please confirm the tuition fees and the available grants and scholarships for international students in the MSc AI program?

All the best,

Sophia

Example (2/2)

Generated draft — Fees and scholarships

Dear Sophia,

Regarding the tuition fees, please consult the official UPM page for the most accurate and up-to-date information: https://www.upm.es/Estudiantes/Estudios_Titulaciones/Estudios_Master/Matricula.

As for scholarships and grants, the Department of Artificial Intelligence awards the José Cuenca Prizes to newly enrolled students (regardless of nationality) who achieve the highest grades after completing their first semester. Additionally, some research groups affiliated with our professors occasionally offer grants tied to specific research projects. For more general information on available aid, please visit: <https://www.upm.es/Estudiantes/BecasAyudasPremios>.

For specific inquiries regarding financial matters or administrative paperwork, I recommend contacting the UPM administration directly at tramitacion.master.official@upm.es.

Best regards,

Damiano Zanardini